

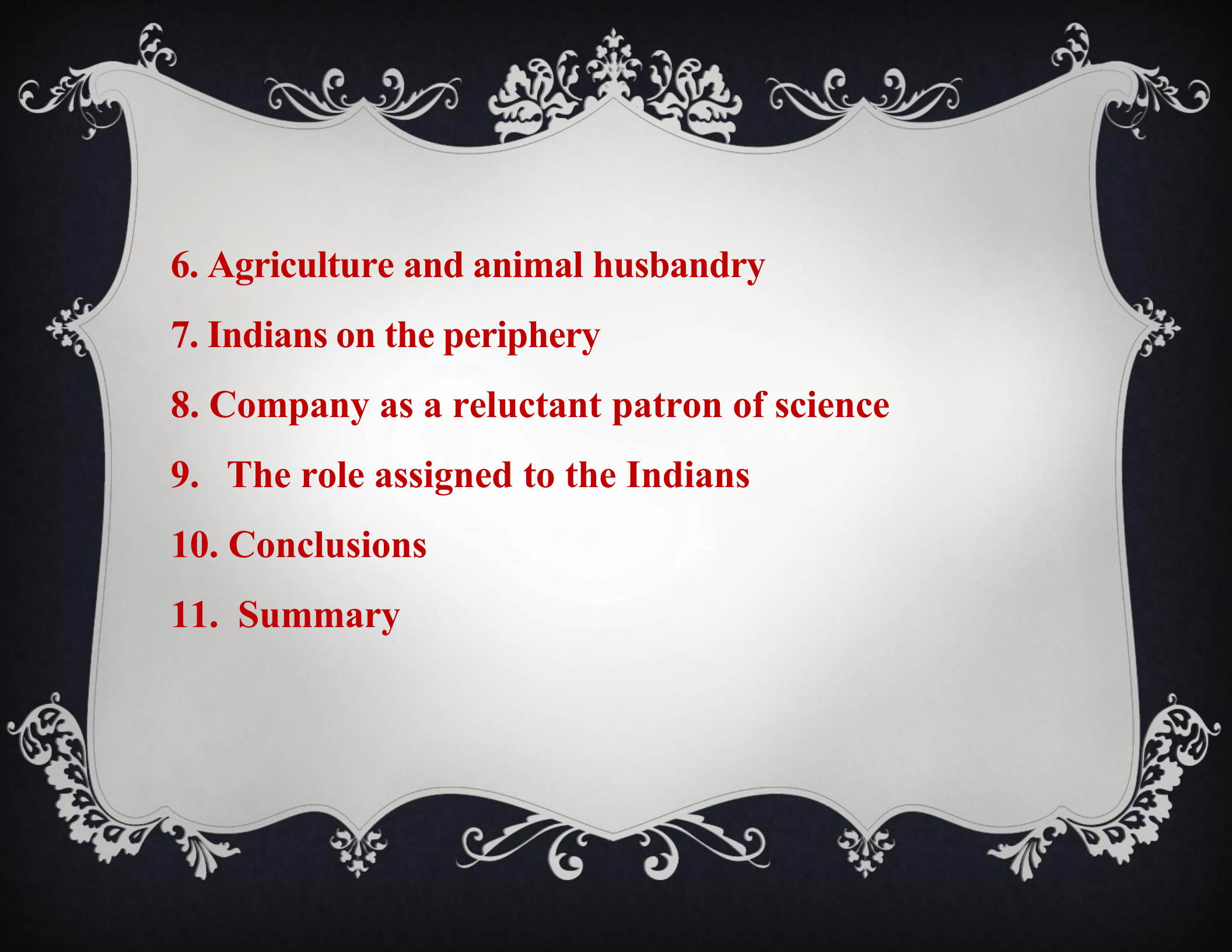
UNIT-III (Chapter-I)
EARLY EUROPEAN SCIENTISTS IN
COLONIAL INDIA

**Surveyors, Botanists, Doctors, Under the
Company's Service**



Objectives

- 1. Introduction**
- 2. Beginning of Modern Science in India**
- 3. Growth of Science- An Instrument of Empire Building**
- 4. Natural History in India during 18th-19th centuries**
- 5. Zoology**

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- 6. Agriculture and animal husbandry**
 - 7. Indians on the periphery**
 - 8. Company as a reluctant patron of science**
 - 9. The role assigned to the Indians**
 - 10. Conclusions**
 - 11. Summary**

Objectives

In this lesson, students investigate introduction modern sciences in India by colonial masters. After studying this lesson you will be able to:

- learn the use of modern science by colonial master in India as a tool for colonization;
- analyze the various European scientist and their work carried out during 18th & 19th century India;.
- investigate development of various branch of natural sciences in colonial India.
- trace the involvement of Indian natives in early scientific endeavor in India by European scientist and
- study the impact of modern science in India upon Indian and their early response.

Introduction

- A major accomplishment of the Renaissance in Europe was the 1498 discovery of the direct sea route to India.
- The great commercial success of the Portuguese spurred the Dutch, the English, and then the French to venture out to the sea.
- The navigational needs of the traders acted as a great incentive for development of science in Europe.
- The best scientists of the time applied their minds to 'discover the longitude'.
- Maritime trade transformed not only European economy but also its state of mind.
- Merchants and artisans now became respectable and influential members of the society.
- The very act of reaching India from Europe showed, a sense of adventure and competence.
- Early European scientific endeavours in India consisted of two disciplines, geography and botany.

Beginning of Modern Science in India

- In the early days, the Europeans- were confined to the coastal areas and had no reason to venture into the interior.
- Geographical exploration therefore fell to the Jesuit priests.
- The first Jesuits arrived in India in 1542 and remained active for more than 200 years.
- In 1759 the King of Portugal banished all Jesuits from his colonies; and in 1773 the Pope banished the order altogether.
- In 1580 Father Monserrate (1536- 1600) on a mission to Emperor Akbar took observations for latitude from Surat to Fatehpur Sikri.
- While geography was left to the missionaries, botany did interest the traders.

- The early use of modern science in India was sporadic and random, and motivated by localized curiosity.
- Additionally, it left the Indians themselves untouched.
- Of far greater significance was the medical expertise of the British doctors. This expertise was sought by the Indian rulers who paid for it in terms of goodwill and trading concessions.
- Gabriel Boughton, a surgeon on the East India Company's ship Hopewell, was sent to Emperor Shahjahan's court at Agra in 1645.
- The surgeon's skills brought the English party into high favour not only with the Emperor but also with his powerful Vazir.

Growth of Science- An Instrument of Empire Building

- With the post-Aurangzeb (1707) collapse of the Mughal Empire, India became available for grabs.
- The 1757 battle of Plassey laid the foundation of the British colonial empire.
- Earlier scientific activity by the Europeans in India had been motivated by commerce and curiosity.
- The chapter will discuss the advent and growth of modern science in India in two stages of development.
- The first stage, called 'the colonial-tool stage', consists of introduction and use of science, especially by the British.
- The second stage, the 'peripheral native stage', came into being when the British were well entrenched in India..

Role of the Portuguese

- The gold coins minted by the Portuguese for use in India depict the armillary sphere, the pre- telescopic basic navigational instrument used for determining the latitude.
- This was Portugal's way of paying tribute to science to which it owed its power.
- The Portuguese arrived in India even before the Mughals did.
- The Portuguese introduced navy as a parameter in India's geo-political equations, placing the Indian rulers at a disadvantage.
- Even when its power was at its peak, the Mughal Empire had to seek the help of the religiously neutral British to thwart the Portuguese attempts at preventing Indian Muslims from sailing to Mecca.
- But by the time the Mughal Empire collapsed, Europe was already on the verge of industrial revolution, so that science and colonialism could feed each other.

French Initiative

- Filling the political vacuum in India required as a first step knowledge about its geography.
- The French were more successful on the scientific front than on the colonial.
- The first worthwhile map of India was compiled in 1752 by the French geographer Jean-Baptiste Bourguignon D'Anville at the request of French East India Company.
- The value of D'Anville's Carte de l'Inde can be judged from the fact that it was reprinted in England in 1754 and then again in 1759, along with the annotated translation of his memoirs

The Trigonometrical Society of India

- The destruction of Tipu in 1799 extended the Company's territories from the east coast to the west.
- The Trigonometrical Survey of Peninsular India was started in 1800 with second-hand instruments bought within the country.
- On 1st January 1818 the survey was renamed the **Great Trigonometrical Survey of India (GTS)** and extended to cover the whole country.
- The GTS came to its own in 1830 under Lt. Col. Sir George Everest (1790- 1866) who was also appointed the Surveyor- General.
- The GTS fixed with great accuracy the longitude and latitude of a large number of places.
- The details were then filled in by the Topographical and Revenue Surveys. In 1878 the three were merged under the name Survey of India.
- Uniformly accurate data from such a huge landmass as India led to the important geodesical theory of isostasy by Archdeacon John Henry Pratt (1808-71) and to a mathematical model of the earth, known as Everest geoids.
- As early as 1787, General William Roy, the founder of the British survey, wrote how desirable it was to determine the length of a degree of latitude on the Coromandel coast and in Bengal. science may be derived from the exact determination of the figure of the Earth, we conceive no other benefit can possibly attend the Admeasurements in Bengal; but that proposed on the Coast of Coromandel will contribute towards the construction of an exact chart of the Coast'.

Astronomical Observatory of Madras

- The **Coromandel coast** is rocky and full of shoals and without a natural port and was a graveyard for the Company's ships.
- A survey of the coast was thus literally a matter of life and death, and eventually in 1785 a professionally trained surveyor-astronomer **Michael Topping (1747-96)** was brought from England, on free passage and equipped with his instruments.
- His work required a reference meridian, an Astronomical Observatory was set up at Madras in 1790. It was the first modern public observatory outside Europe.
- Although the Company had grandiosely declared that the purpose of the Observatory was to '**promote the knowledge of astronomy, geography, and navigation in India**', it was clear that the main aim was to promote the **Company's profitability**.
- Increased sea-trade activities of the British required familiarity with the southern skies. In 1844 after 14 years of labour, **Thomas Glanville Taylor (1804-48)** produced the celebrated Madras catalogue giving positions of about 11,000 southern stars.
- India was ideal for extensive photography of the sun, which was not possible in cloudy Britain. Also, it was then believed that a study of the sun will help predict the failure of the monsoons.

- In 1878 solar photography was started at Survey of India, Dehra Dun, and photographs were sent to England for analysis. A solar observatory was established by the Imperial Government at Kodaikanal in 1899 (which still exists).
- Once the Trigonometrical Survey was begun, the Government lost interest in Madras Observatory. All surveys were manned by military officers.
- In 1834, on orders from the Government, astronomical instruments from the Survey were issued to enable the former Bombay Astronomer to observe the phenomenon of the opposition of Mars.
- In 1818 **Dr Henry Voysey** a surgeon who doubled as a geologist was attached to the GTS so that he could draw 'attention to anything that might influence geometrical and astronomical observations'.
- Voysey's reports included one on the stone used in building the Taj at Agra. He also reported on diamond mines of South India.
- Industrial revolution meant the realization that coal was more important than diamond. As the steamer ships were pressed into use, the Government became interested in coal fields. This led to the appointment of a Geological Surveyor to the Company, and in 1851 to the Geological Survey of India.
- Geological evidence in support of the continental-drift hypothesis came from India; this fact is commemorated in the name 'Gondwana' for the ancient southern super-continent. The name, Gondwana System, was introduced in 1872 by Henry Benedict Medlicott (1829-1905), from 1854 professor of geology at Roorkee and later superintendent of the Geological Survey of India.

Early Botanical Study

- Systematic study of botany in India was pioneered by John Gerald Koenig, a native of the Baltic province of Courland, who came to the Danish settlement at Tranquebar near Madras.
- A recognized centre for botanical activity was provided by the East India Company in July 1787 with the establishment of a 300-acre botanical garden at Sibpur on the bank of Hooghly, near Calcutta.
- At the fall of Mysore, the botanical garden at Bangalore (the Lal Bagh) was appropriated by the Company 'as a depository for useful plants sent from different parts of the country'. The Company's botanist at Madras (Dr Benjamin Heyne) was ordered by the Governor-General to accompany the Surveyor.
- An important task assigned to the colonial botanical gardens was dealing with malaria, the scourge of the tropics and the biggest obstacle to colonial expansion. The bark of the cinchona tree as a cure for malaria was introduced in Europe from South America in 1640 by a Spanish lady Countess del Cinchon.
- In 1820, two French chemists Pierre Joseph Pelletier and Joseph Bienaime Conventou succeeded in extracting the alkaloid of quinine from cinchona bark, and commercial production of quinine began in 1827.
- Until the 1850s all the world's cinchona bark came from the forests of Peru, Bolivia, Ecuador, and Colombia, where the trees grew wild.
- Intense experimentation at the Indian botanical gardens made India self-sufficient in quinine. Almora-born British physician Ronald Ross (1857-1932) discovered in 1897 that the germ of malaria is carried by Anopheles mosquito, and received the 1911 Nobel prize.
- Later, in 1904 Ross served as a consultant at Panama where the long-needed canal could finally be built only when malaria was eradicated.

Natural History in India during 18th-19th centuries

- European access to India was a multi-dimensional phenomenon.
- The merchant-rulers were keen to identify commodities that could be profitably exported to Europe, cultivate commercial plants in India that grew outside their possessions, and find substitutes for drugs and simples that were obtained from the Americas.
- On their part, the Christianity enthusiasts in Europe viewed European rule in India as a godsend for propagating the Gospel in the east. These seemingly diverse interests converged at various levels.
- Christian missionaries as a body were the first educated Europeans in India. They constituted a valuable resource for naturalists in Europe. European interest in their field work brought them scientific recognition as well as the much needed cash. More significantly, they introduced the colonial administrators, especially the medical men, to systematic botany.
- Europe was introduced to western Indian drugs and simples by the 1563 work of the Goa-based Portuguese physician Garcia d'Orta (1501-1568).
- A century later, during 1678-1693, the Dutch administrator Hendrik Adriaan (1636-1691) furnished Europe with information from south India.
- Now, individual European naturalists and institutions wanted not only specimens from India but also bits of tacit knowledge resident in local population.

Zoology

- Zoology was a late starter in British India, because animals had no commercial value.
- Academic studies could not take place early because of the known aversion of Sir William Jones, the founder of the Asiatic Society.
- In 1796, two years after Jones' death, the Society announced their intention of establishing a natural history museum and invited donations.
- In 1841 Edward Blyth (1810-1873) was brought from England as the curator. In spite of drudgery of his routine work and unpleasant disputes within his employers, Blyth built up the museum and emerged as an authority on Indian and Burmese birds.
- In 1888, he was described as the **Father of Indian Ornithology**.
- Blyth arranged to supply to England animals which were in demand there for private and public zoos. Similarly he imported live animals into India; 'Natives of enormous wealth are the purchasers, who care not what they give for what they particularly fancy'.
- Thanks to Blyth's official exertions, the Museum built an impressive collection.
- John Anderson took over as the Curator in 1865. Next year he became the superintendent of Indian Museum which was founded using the Asiatic Society Museum collection as its nucleus.
- In 1891 Botanical Survey of India was constituted but by this time the best in colonial field science was already over.
- Further, in 1916, the anthropological and zoological sections of the Museum were made into Zoological Survey of India, a step of administrative significance rather than scientific.

Agriculture and animal husbandry

- Farming and animal husbandry were the two most important economic activities in India.
- In agriculture, it was interested only in exports, but for administrative reasons it wished to avoid famines.
- From among the animals, for a long time, its attention was focused on the horse which was required for service in the military.
- An army stud department was established in Bengal as early as 1779, and a horse breeding farm was established in 1784 in a vast 1350-acre estate near a village called **Pusa in Bihar**. The farm was closed in 1874.
- The pre-existing **Amrit Mahal cattle breeding establishment in Mysore** was taken over by Madras government in 1813 for research on pack animals for the army.
- Agriculture was a low priority area for colonial India. The British interest in Indian agriculture primarily focused on industrial cash crops.
- An agricultural department was established in 1871 'chiefly in relation to the supply of cotton from India'. The department was however declared to be a failure and closed in 1878.
- A central agriculture department was set up afresh in 1881.
- Agricultural research began as late as 1905 on the personal initiative of the Viceroy Lord Curzon, when an institute was established in the **Pusa estate**.

Indians on the periphery

- Collection of plants from the field, preparation of specimens and drawing their sketches were important parts of a European naturalist's project.
- To retain ownership of drawings and specimens, the part-time naturalists preferred to pay their staff from their own pocket rather than bill the Company, as was done by Buchanan and Roxburgh to their disadvantage.
- To protect the specimens and the paper on which they were preserved from attacks by the ants, the feet of the wooden cabinets were kept immersed in troughs of water.
- To replenish the water lost by evaporation, a person was hired whose job was to keep the troughs filled with water 'until the shadows of evening came on'.
- Calcutta Garden did have artists on its staff. Many Europeans did not bother to record the names of the artists working for them.
- To our frustration, but not surprisingly, none of the Indians so employed seems to have left behind any account.
- In his 27 years of career in Madras Presidency, Wight employed two Telugu artists. Rungiah worked from Wight's arrival in 1826 till 1845 and the former's pupil Govindoo from 1845 till Wight's retirement in 1853.
- Rungiah was taught the use of microscope by William Griffith.
- Rungiah might have served Wight's predecessors as well, and Govindoo continued to work post-Wight at Madras for Hugh Cleghorn and Richard Henry Beddone (1830-1911).

- The artists in North India came from the royal court tradition rather than the temple.
- A painter notable for his versatility, is **Lutchman Singh or Lakshman Singh** who was employed in Calcutta garden. He painted a portrait of George Potter in the 'Murshidabad style' in 1828, when Wallich was away to Europe.
- His services were borrowed by Royle in Saharanpur where he made three zoological drawings. Interestingly, Lachhman Singh 'decamped for a spell to work as a Court painter in one of the Punjab hill states'.
- But, once the new art of lithography was introduced, talented persons who had no previous connection with painting could become professionals. This was part of a broader phenomenon.
- The traditional caste profession equation underwent a significant transformation. Old professions which were learnt through apprenticeship remained the preserve of the associated caste as before.
- But if a professions was taught in the educational institutions, it got decoupled from caste.
- Newly introduced professions by definition had no caste connotation and could be adopted by any one.

Company as a reluctant patron of science

- Just as evangelicals in Britain wanted the Company's help in spreading the Gospel in its territories, the British men of science wanted the Company to lend its support in advancing science. Both were disappointed; the former fully, the latter partially.
- The Company knew the value of science and pressed it into service whenever the need arose. It however was extremely reluctant to support science for its own sake.
- Yet the Company though powerful had its limitations and compulsions. It was answerable to the parliament, subject to government control and sensitive to home public opinion.
- As science progressed in Europe and scientific community became visible and influential the merchant rulers were forced to pay homage to science, but only to a point.
- Yielding to appeals by the scientific community, the Company permitted **Hugh Falconer** to remain on full salary in London from 1844 till 1847 to work on the **Shivalik fossils**, but then brought him back under threat of forfeiture of pension.
- The work was finally published in the post-Company period, in 1868. Similarly, in 1851 notwithstanding a memorandum from the British Association, the Company refused to promote a **project on Flora Indica by Joseph Dalton Hooker** and his collaborator Thomas Thomson (1817-78), later the superintendent of **Calcutta Garden 1854-61**.

The role assigned to the Indians

- The missionaries interacted with the Indians and documented their tacit knowledge with a view to incorporating it into the European mainstream.
- Indians by and large remained nameless and faceless attendants in the European club of science. Whenever an Indian served particularly well credit went to his master and mentor for his success in improving the natives. There were rare exceptions to this general rule.
- In 1843 George Everest successfully interceded with the Court of Directors to obtain for his gifted Indian mathematical instrument maker **Syed Mir Mohsin Hussain** the same designation if not the same salary as his European predecessor.
- When the Indian explorer **Nain Singh** was awarded a gold watch by the Royal Geographical Society, he was almost irrelevant at the award ceremony held in England on 25 May 1868. There was no question of his being invited to the ceremony; he was not even mentioned by name.
- '**Pundit employed by Captain Montgomerie**', who was the real hero as far as the Geographical Society was concerned.
- In the same vein **Falconer's 1868 biographer Murchison** smugly noted that 'the intelligence, docility, and exquisite manual dexterity of the natives' when 'backed by their faith in the guiding head of the European' 'furnished an inexhaustible fund of resource'.
- Probably the most trivial duty assigned to Indians in the cause of natural history was at Botanic Garden Calcutta. To prevent the ants from eating the specimens and the paper on which they were glued the feet of the cabinet were kept immersed in troughs of water.
- To replenish the water lost by evaporation, Wallich hired a person to keep the troughs filled with water until the shadows of evening came on and relieved him from his tedious and monotonous task.
- The court tradition in north India and the temple tradition in south India were combined with European requirements to give rise to a unique style which is of interest not only to students of natural history but also art history.

Conclusion

- Early use of modern science in India was sporadic and desultory, and motivated by localized curiosity.
- Most of it had no contemporaneous significance and was incorporated into the main body of science much later.
- Additionally, it left the Indians themselves untouched.
- The British-sponsored science by the very reason of its existence was field science. Geography, geodesy, geology, botany, zoology, paleontology, archaeology, medicine and even astronomy-all these stemmed from the physical and cultural novelty of India.
- British rulers were not interested in science in India for its own sake, but in using it to further their interests.
- Whenever their practical needs pointed a finger towards a particular branch of science, attention was paid to that science.
- Astronomy was the first modern science to be brought to India, as a geographical and navigational aid.
- The missionaries interacted with the Indians and documented their tacit knowledge with a view to incorporating it into the European mainstream.
- Indians by and large remained nameless and faceless attendants in the European club of science.

Summary

- *Early use of modern science in India was sporadic and desultory, and motivated by localized curiosity. Most of it had no contemporaneous significance and was incorporated into the main body of science much later. Additionally, it left the Indians themselves untouched.*
- *British rulers were not interested in science in India for its own sake, but in using it to further their interests. Whenever their practical needs pointed a finger towards a particular branch of science, attention was paid to that science.*
- *The 1757 battle of Plassey laid the foundation of the British colonial empire and also science was now pressed into the cause of empire-building and institutionalized.*
- *Astronomy was the first modern science to be brought to India, as a geographical and navigational aid. Its early use was however sporadic and mostly out of personal curiosity.*
- *Surveys were continually required for military purposes. Geographical location of important places in the country was determined with alacrity by „borrowing a sextant here, a watch there, and a quadrant in another quarter, from different officers at Calcutta who happened to possess them”.*
- *Systematic study of botany in India was pioneered by John Gerald Koenig, a native of the Baltic province of Courland, who came to the Danish settlement at Tranquebar near Madras. In 1891 Botanical Survey of India was constituted.*

- *The British desire for exploration and increased revenue led to the epoch- making discovery of fossil fauna in the Shivalik hills.*
- *European access to India was a multi-dimensional phenomenon. Europe was excited about the opportunities that the vast landmass of India offered in natural history studies. European interest in their field work brought them scientific recognition as well as the much needed cash. More significantly, they introduced the colonial administrators, especially the medical men, to systematic botany.*
- *Farming and animal husbandry were the two most important economic activities in India. The colonial government''s interests in them were however selective and self-centred. In agriculture, it was interested only in exports, but for administrative reasons it wished to avoid famines. From among the animals, for a long time, its attention was focused on the horse which was required for service in the military.*
- *Physics never became a part of colonial science in India. A small exception had to be made in the case of chemistry because it was required as an administrative aid. A fairly well equipped chemistry laboratory was set up in the Presidency College Calcutta in 1874.*
- *In the early phase a lot of Indian were used as helper by European scientist, which result in the involvement of Indian natives in the world of modern science.*

Thank you